

ORF309 S2026 Midterm1

March 2026

Problem 1. You roll two fair six-sided dice and tell me the product of the two numbers you get.

(a) [10 points] What is a good sample space for this random experiment? How to define the corresponding probability measure on it?

(b) [10 points] What is the expectation of the number you tell me?

Solution. (a) A good sample space Ω represents all possible outcomes of rolling two distinguishable six-sided dice.

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}$$

This sample space contains exactly $6 \times 6 = 36$ equally likely outcomes, because the dice are fair. We can define the corresponding probability measure $\mathbb{P}$ as a discrete uniform distribution over Ω :

$$\mathbb{P}(\{(i, j)\}) = \frac{1}{36}, \quad \forall (i, j) \in \Omega.$$

As a result, for any event $A \subset \Omega$:

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{|A|}{36}.$$

(b) Let X be the outcome of the first die and Y be the outcome of the second die. We are looking for the expectation of their product, $\mathbb{E}[XY]$. Since the dice rolls are independent, the expectation of their product is the product of their expectations:

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

For a single fair six-sided die, the expected value is:

$$\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

Therefore, the expected value of the product is:

$$\mathbb{E}[XY] = 3.5 \times 3.5 = 12.25.$$

□

Remark. The solution presented here is the most standard way of designing the sample space: it includes all the information needed to describe the whole random experiment. The product of the two numbers can then be represented as a function defined on the sample space (i.e. a random variable).

It is acceptable to directly define Ω to be the possible outcomes of the product:

$$\Omega = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 18, 20, 24, 25, 30, 36\}.$$

But one must recognize that these outcomes are **not** of equal probabilities:

$$\begin{aligned}\mathbb{P}(\{1\}) &= \mathbb{P}(\{9\}) = \mathbb{P}(\{16\}) = \mathbb{P}(\{25\}) = \mathbb{P}(\{36\}) = \frac{1}{36}, \\ \mathbb{P}(\{2\}) &= \mathbb{P}(\{3\}) = \mathbb{P}(\{5\}) = \mathbb{P}(\{8\}) = \mathbb{P}(\{10\}) = \mathbb{P}(\{15\}) = \mathbb{P}(\{18\}) = \mathbb{P}(\{20\}) = \mathbb{P}(\{24\}) = \mathbb{P}(\{30\}) = \frac{2}{36}, \\ \mathbb{P}(\{4\}) &= \frac{3}{36}, \quad \mathbb{P}(\{6\}) = \mathbb{P}(\{12\}) = \frac{4}{36}.\end{aligned}$$

In addition, the independence structure cannot be recovered from this reduced form of sample space.

Problem 2. Researchers have observed that during the winter, students' willingness to attend morning lectures depends on weather conditions. On a day without snowfall, each student attends the lecture independently with probability $p_1 = 0.7$. On a snowy day, each student attends independently with probability $p_2 = 0.1$. On any given winter day, snow occurs with probability $q = 0.2$.

Assume there are 100 students enrolled in the class, and that attendance decisions are mutually independent across students, conditional on the weather.

(a) [10 points] What is the expected number of students attending the lecture on a random winter day?

(b) [10 points] Suppose that on a particular observed day, exactly 30 students attended the lecture. Given this information, what is the probability that it snowed on that day?

Solution. (a) Let N be the number of students attending. We can use the Law of Total Expectation. Let S represent the event that it snows, and S^c represent the event that it does not snow.

$$\mathbb{E}[N] = \mathbb{E}[N \mid S]\mathbb{P}(S) + \mathbb{E}[N \mid S^c]\mathbb{P}(S^c)$$

Given the conditions, the expected number of students for each weather type is simply the total number of students multiplied by the individual probability of attending:

- $\mathbb{E}[N \mid S] = 100 \times 0.1 = 10$
- $\mathbb{E}[N \mid S^c] = 100 \times 0.7 = 70$

Now, plugging in the probabilities $\mathbb{P}(S) = 0.2$ and $\mathbb{P}(S^c) = 0.8$:

$$\mathbb{E}[N] = (10)(0.2) + (70)(0.8) = 2 + 56 = 58$$

The expected number of students attending is **58**.

(b) We need to find $\mathbb{P}(S \mid N = 30)$. We will use Bayes' Theorem:

$$\mathbb{P}(S \mid N = 30) = \frac{\mathbb{P}(N = 30 \mid S)\mathbb{P}(S)}{\mathbb{P}(N = 30 \mid S)\mathbb{P}(S) + \mathbb{P}(N = 30 \mid S^c)\mathbb{P}(S^c)}$$

Conditional on the weather, the number of attending students follows a Binomial distribution:

- $\mathbb{P}(N = 30 \mid S) = \binom{100}{30}(0.1)^{30}(0.9)^{70}$
- $\mathbb{P}(N = 30 \mid S^c) = \binom{100}{30}(0.7)^{30}(0.3)^{70}$

Substituting these into Bayes' Theorem (and canceling the $\binom{100}{30}$ terms):

$$\mathbb{P}(S \mid N = 30) = \frac{0.2(0.1)^{30}(0.9)^{70}}{0.2(0.1)^{30}(0.9)^{70} + 0.8(0.7)^{30}(0.3)^{70}}.$$

□

Problem 3. Alice and Bob play a game involving a fair six-sided die and a fair coin. In each round, Alice rolls the die and Bob flips the coin. Alice earns \$3 each time she rolls a 6, and Bob earns \$1 each time he flips a head. All outcomes are assumed to be mutually independent across players and across rounds.

(a) [10 points] If the game lasts for 10 rounds, compare the variances of their total earnings.

(b) [10 points] If the game lasts until Alice earns \$3, what is the variance of Bob's total earning?

You may use the following fact from Calculus without justification:

$$\sum_{n=1}^{\infty} n^2 p(1-p)^{n-1} = \frac{2-p}{p^2}, \quad \forall p \in (0, 1).$$

Solution. (a) Let's first find the variance of earnings in a single round.

- **Alice:** Let A_i be Alice's earnings in round i . $A_i = 3$ with probability $1/6$, and 0 with probability $5/6$.

$$\mathbb{E}[A_i] = 3 \times \frac{1}{6} = 0.5$$

$$\mathbb{E}[A_i^2] = 3^2 \times \frac{1}{6} = \frac{9}{6} = 1.5$$

$$\text{Var}(A_i) = \mathbb{E}[A_i^2] - (\mathbb{E}[A_i])^2 = 1.5 - (0.5)^2 = 1.5 - 0.25 = 1.25$$

- **Bob:** Let B_i be Bob's earnings in round i . $B_i = 1$ with probability $1/2$, and 0 with probability $1/2$.

$$\text{Var}(B_i) = p(1-p) = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = 0.25$$

Since the 10 rounds are independent, the variance of the total earnings is the sum of the variances.

- $\text{Var}(A_{\text{total}}) = 10 \times 1.25 = 12.5$

- $\text{Var}(B_{total}) = 10 \times 0.25 = 2.5$

Comparison: Alice's variance (12.5) is exactly 5 times Bob's variance (2.5).

(b) Let R be the number of rounds until Alice earns \$3 by rolling a 6. R follows a Geometric distribution with success probability $p = 1/6$. Let B_i be Bob's earning in round i , where $B_i = 1$ if he flips a head and 0 otherwise. Thus, $B_i \sim \text{Bernoulli}(0.5)$. Bob's total earning is the random sum $Y = \sum_{i=1}^R B_i$.

To find $\text{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2$, we first find the first moment by conditioning on R :

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|R]] = \mathbb{E}[0.5R] = 0.5(6) = 3.$$

For the second moment $\mathbb{E}[Y^2]$, we condition on the event $\{R = n\}$. Given $R = n$, the sum Y is a Binomial random variable with parameters n and 0.5:

$$\mathbb{E}[Y^2|R = n] = \text{Var}(Y|R = n) + (\mathbb{E}[Y|R = n])^2 = n(0.5)(0.5) + (0.5n)^2 = 0.25n + 0.25n^2.$$

By the Law of Total Expectation, we integrate this over all possible values of R :

$$\mathbb{E}[Y^2] = \mathbb{E}[0.25R + 0.25R^2] = 0.25\mathbb{E}[R] + 0.25\mathbb{E}[R^2]$$

Using the provided calculus fact, we find the second moment of R :

$$\mathbb{E}[R^2] = \sum_{n=0}^{\infty} n^2 p (1-p)^{n-1} = \frac{2 - 1/6}{(1/6)^2} = \frac{11/6}{1/36} = 66.$$

Substituting $\mathbb{E}[R] = 6$ and $\mathbb{E}[R^2] = 66$ into our expression for $\mathbb{E}[Y^2]$:

$$\mathbb{E}[Y^2] = 0.25(6) + 0.25(66) = 1.5 + 16.5 = 18.$$

Finally, the variance is:

$$\text{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = 18 - (3)^2 = 9.$$

□